

Translation operations on trianguline (φ, Γ) -modules

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Background

Notation

$E/K/\mathbb{Q}_p$ finite extensions, $\mathbf{G} := \mathrm{GL}_n$, $\mathbf{G} \supset \mathbf{B} \supset \mathbf{T}$ Borel subgroup and max'l torus, $\mathfrak{g} \supset \mathfrak{b} \supset \mathfrak{t}$ Lie algebras, $G := \mathbf{G}(K)$, $B := \mathbf{B}(K)$, $T := \mathbf{T}(K)$.
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Locally analytic p -adic Langlands correspondence

$$\left\{ \begin{array}{l} n\text{-dim'l continuous } E\text{-linear} \\ \text{representations of } \mathrm{Gal}_K \end{array} \right\} \xleftrightarrow{\quad ? \quad} \left\{ \begin{array}{l} \text{some locally } \mathbb{Q}_p\text{-analytic} \\ \text{representations of } \mathrm{GL}_n(K) \end{array} \right\}$$

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 $X_{\hat{K}_\infty^b} := (\text{Spa}(W(\mathcal{O}_{\hat{K}_\infty^b})) \setminus \{p[\varpi^b] = 0\})/\varphi^{\mathbb{Z}}$, $[\infty]$ = the untilt $\hat{K}_\infty$ of $\hat{K}_\infty^b$

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expected to satisfy **compatibility** between Hodge-Tate(-Sen) weights and infinitesimal characters via $\mathcal{Z}(\mathfrak{g}) \xrightarrow[\simeq]{\text{Harish-Chandra}} \mathcal{O}(\mathfrak{t}^*)^W$; namely,

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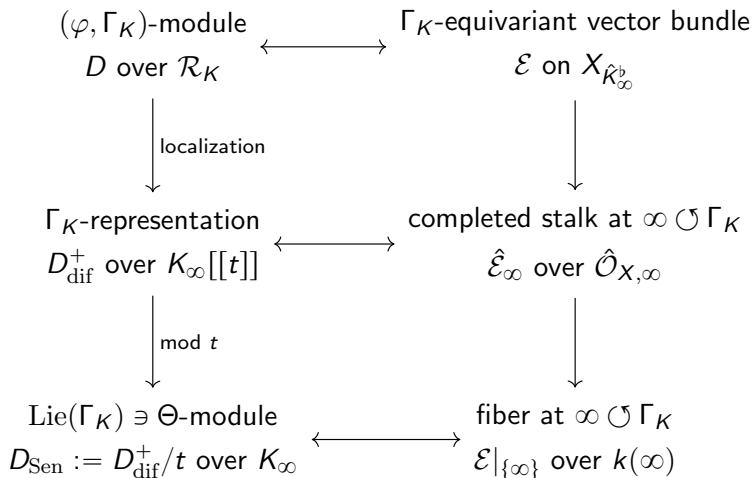
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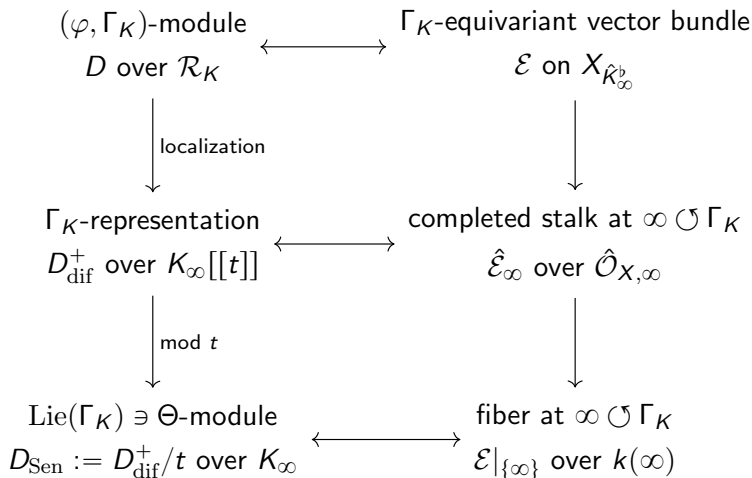
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$$\mathbf{h} = (h_1, \dots, h_n) \mapsto \chi_{\lambda_{\mathbf{h}}}, \quad \lambda_{\mathbf{h}} := \mathbf{h} - \rho.$$

B_{dR}^+ -Sen theory on Fargues-Fontaine curve



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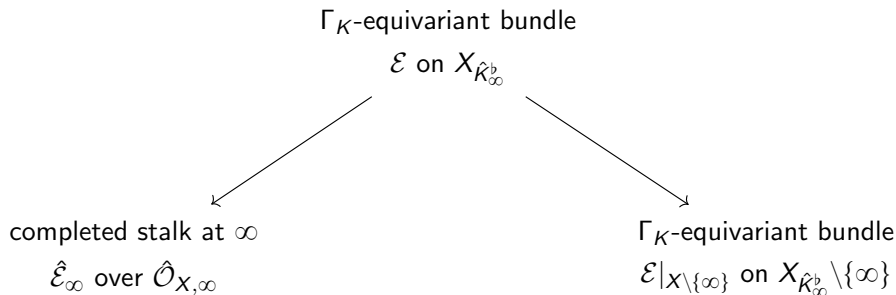


$\rightsquigarrow$ Sen polynomial $P_{\text{Sen}}(T) := \text{char.poly}(\Theta) \in K[T]$

$\rightsquigarrow$ (Hodge-Tate-)Sen weights $\text{wt}(D) := \text{zeros}(P_{\text{Sen}}(T))$.

Beauville-Laszlo and Change of weights

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Γ_K -equivariant bundle

$\mathcal{E}$ on $X_{\hat{K}_\infty^b}$

completed stalk at ∞
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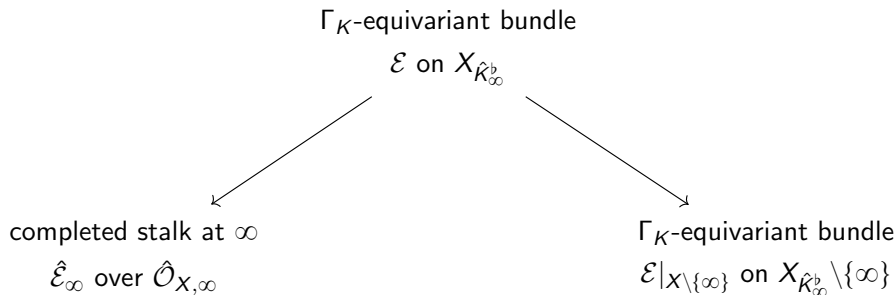
Γ_K -equivariant bundle
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Beauville-Laszlo $\rightsquigarrow D \mapsto (D_{\text{dif}}^+, D[1/t])$ defines an equivalence.

“Change of weights” = Γ_K -equivariant modification at ∞ of the lattice D_{dif}^+ inside $D[1/t] \otimes_{\mathcal{R}[1/t]} K_\infty((t))$.

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Work with coefficients in affinoid E -algebras A .

Change of weights for trianguline (φ, Γ) -modules

If D comes with a B -structure $0 \subsetneq \mathrm{Fil}^1(D) \subsetneq \cdots \subsetneq \mathrm{Fil}^n(D) = D$ with graded pieces $\mathrm{Fil}^i(D)/\mathrm{Fil}^{i+1}(D) \cong \mathcal{R}(\delta_i)$ for continuous $\delta_i : K^\times \rightarrow A^\times$, then $\mathrm{wt}(D)$ are given by derivatives $\mathrm{wt}(\delta_i) = \left. \frac{d}{d\gamma}(\delta_i) \right|_{\gamma=1}$ of the δ_i .

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We may view $D \in \mathrm{Ext}_{\varphi, \Gamma}^1(Q_i, \mathrm{Fil}^i(D))$ and pullback along $t^k Q_i \subset Q_i$, where Q_i denotes the quotient (φ, Γ) -module $D/\mathrm{Fil}^i(D)$, e.g.,

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- Natural questions: (when) is it independent of the choice of B -structure, and (when) is it invertible?

Using Sen polynomials

Proposition (Wu, 2024)

If $P_{\text{Sen},D}(T) = Q(T)S(T)$ with coprime monic $Q, S \in A[T]$, there exists unique (φ, Γ) -submodule $tD \subset D' \subset D$ with $P_{\text{Sen},D'}(T) = Q(T-1)S(T)$.

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Consequences. If we consider those trianguline (φ, Γ) -modules on $\mathrm{Sp}(A)$ that are “**weight-uniform**” at all geometric points (all B -structures there induce the same ordering of Sen weights, e.g., *non-critical crystabelline*),

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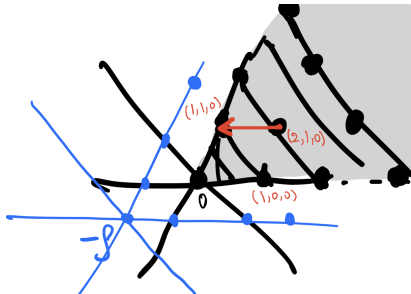
$\rightsquigarrow$ the pullback map $p_{\mathbf{k}} : \mathfrak{X}_B^{\mathrm{wu}} \rightarrow \mathfrak{X}_B^{\mathrm{wu}}$ descends to an isomorphism of certain substacks $p_{\mathbf{k}} : \mathfrak{X}_n^{\mathrm{wu}, \mathbf{k}} \xrightarrow{\cong} \mathfrak{X}_n^{\mathrm{wu}, -\mathbf{k}}$ of the moduli stack $\mathfrak{X}_n$ of rank n Γ_K -equivariant vector bundles on the Fargues-Fontaine curve $X_{\hat{K}_{\infty}^b}$.

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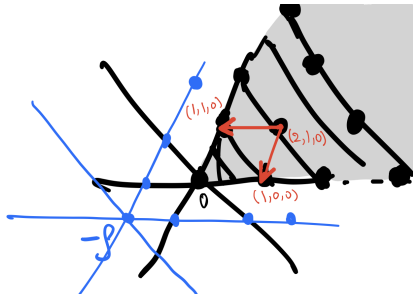


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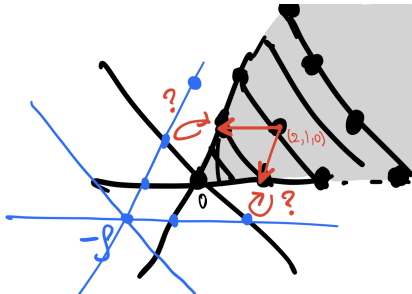


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Compatibility under Ding's crystabelline correspondence

For non-critical crystabelline (φ, Γ) -modules D over $\mathcal{R}_{K,E}$ of regular weights $\mathbf{h} = (h_1 > h_2 > \cdots > h_n) \in \mathbb{Z}^n$, with smooth characters $(\phi_i : \mathbb{Q}_p^\times \rightarrow E^\times)$ such that $D[1/t] \cong \bigoplus_{i=1}^n \mathcal{R}_{\mathbb{Q}_p, E}(\phi_i)[1/t]$, Ding has constructed a 1-1 correspondence:

Theorem (Ding, 2024)

For $D \in \Phi\Gamma_{\text{nc}}(\phi, \mathbf{h})$, there is an explicit locally analytic representation $\pi_1(D)$ of $\text{GL}_n(\mathbb{Q}_p)$ that *completely determines* D , in the sense that if $D, D' \in \Phi\Gamma_{\text{nc}}(\phi, \mathbf{h})$, then $D \cong D'$ precisely when $\pi_1(D) \cong \pi_1(D')$.

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From the work of Ding, Jena-Lahiri-Strauch, and Wu, it follows that

Theorem

If two regular weights $\mathbf{h} \rightsquigarrow \mathbf{h}'$ are related by $p_{\mathbf{k}}$, then for $D \in \Phi\Gamma_{\text{nc}}(\phi, \mathbf{h})$,

$$T_{\lambda_{\mathbf{h}}}^{\lambda_{\mathbf{h}'}}(\pi_1(D)) = \pi_1(p_{\mathbf{k}}(D)).$$

Compatibility under Ding's crystabelline correspondence

Remarks. (1) The construction $\pi_1(D)$ by Ding was done for any finite extension $K/\mathbb{Q}_p$. The compatibility between $\mathbf{h} \rightsquigarrow \mathbf{h}'$ (not realizable by $\rho_{\mathbf{k}}$) and translation functor $T_{\lambda_{\mathbf{h}}}^{\lambda_{\mathbf{h}'}}$, for any $K/\mathbb{Q}_p$, still hold.

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(2) For integral weights $\mathbf{h}$, the (φ, Γ) -modules D are “almost de Rham”. Wu described the completion at Sen weights $\mathbf{h}$ of the moduli stack

$$(\mathfrak{X}_n)_{\mathbf{h}}^{\wedge} \simeq (\mathfrak{X}_n)_0^{\wedge} \times_{\mathfrak{g}/\mathrm{GL}_n} \widetilde{\mathfrak{g}}_{P_{\mathbf{h}}}/\mathrm{GL}_n,$$

via local model, and thus obtained geometric “change of weights” maps:

$$f_{\mathbf{h}, \mathbf{h}'} : (\mathfrak{X}_n)_{\mathbf{h}}^{\wedge} \simeq (\mathfrak{X}_n)_0^{\wedge} \times_{\mathfrak{g}/\mathrm{GL}_n} \widetilde{\mathfrak{g}}_{P_{\mathbf{h}}}/\mathrm{GL}_n \longrightarrow (\mathfrak{X}_n)_0^{\wedge} \times_{\mathfrak{g}/\mathrm{GL}_n} \widetilde{\mathfrak{g}}_{P_{\mathbf{h}'}}/\mathrm{GL}_n \simeq (\mathfrak{X}_n)_{\mathbf{h}'}^{\wedge}$$

$$\text{So, } T_{\lambda_{\mathbf{h}}}^{\lambda_{\mathbf{h}'}}(\pi_1(D)) = \pi_1(f_{\mathbf{h}, \mathbf{h}'}(D)).$$

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(2) For integral weights $\mathbf{h}$, the (φ, Γ) -modules D are “almost de Rham”. Wu described the completion at Sen weights $\mathbf{h}$ of the moduli stack

$$(\mathfrak{X}_n)_{\mathbf{h}}^{\wedge} \simeq (\mathfrak{X}_n)_0^{\wedge} \times_{\mathfrak{g}/\mathrm{GL}_n} \widetilde{\mathfrak{g}}_{P_{\mathbf{h}}}/\mathrm{GL}_n,$$

via local model, and thus obtained geometric “change of weights” maps:

$$f_{\mathbf{h}, \mathbf{h}'} : (\mathfrak{X}_n)_{\mathbf{h}}^{\wedge} \simeq (\mathfrak{X}_n)_0^{\wedge} \times_{\mathfrak{g}/\mathrm{GL}_n} \widetilde{\mathfrak{g}}_{P_{\mathbf{h}}}/\mathrm{GL}_n \longrightarrow (\mathfrak{X}_n)_0^{\wedge} \times_{\mathfrak{g}/\mathrm{GL}_n} \widetilde{\mathfrak{g}}_{P_{\mathbf{h}'}}/\mathrm{GL}_n \simeq (\mathfrak{X}_n)_{\mathbf{h}'}^{\wedge}$$

So, $T_{\lambda_{\mathbf{h}}}^{\lambda_{\mathbf{h}'}}(\pi_1(D)) = \pi_1(f_{\mathbf{h}, \mathbf{h}'}(D))$. Cf. Wu's paper for speculations on translation operations via categorical p -adic Langlands correspondence.

Thank you!



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